

PHYSICS 101 · ANSWER KEY - PROFESSOR COPY

Homework 2: Light

Grading guide. Total 21 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — The rainbow

1. Write the colors of the rainbow in order from the LONGEST waves to the SHORTEST. Then name the invisible color just past each end. (3 pts)

Answer: Red, orange, yellow, green, blue, violet (2 pts; 1 pt if one swap). Past red: infrared; past violet: ultraviolet (1 pt).

2. Draw Newton's experiment: a beam of white light entering a prism and fanning out into colors on the other side. Label the red end and the violet end. (2 pts)

Answer: A beam in, a triangle (prism), a fan of colors out (1 pt); red and violet labelled at the two edges of the fan (1 pt). Any order of colors between them is fine as long as red and violet are the outside edges.

Part B — Why things have color

3. For each object, write which color(s) bounce back to your eye and what happens to the rest. (4 pts)

Answer: Tomato: red bounces back; the rest are absorbed. Leaf: green bounces; the rest absorbed. White paper: all colors bounce back; none absorbed. Black shirt: nothing bounces back; all absorbed (and it gets warm). 1 pt per row.

Part C — The two-color trick

4. In your own words, explain how the pulse oximeter turns two colors of light into an oxygen percentage. Use the words red, infrared, and hemoglobin. (3 pts)

Answer: Red and infrared light shine through the foot to a detector (1 pt). Hemoglobin loaded with oxygen lets red through but absorbs infrared; empty hemoglobin absorbs red and lets infrared through (1 pt). Comparing how much of each color gets through tells the machine what fraction of the hemoglobin is loaded — the SpO_2 percentage (1 pt).

5. Why does the machine use only the PULSING part of the light signal? (1 pt)

Answer: A. Only the fresh blood in the arteries pulses; skin, bone, and vein blood don't. - Aoyagi's insight (1972): throw away the steady part and keep the heartbeat wiggle, which comes only from arterial blood.

Part D — The lamp that cures

6. Why blue? Explain what blue light does to a bilirubin molecule and why that helps a baby whose liver isn't ready. (2 pts)

Answer: Bilirubin is yellow, so it absorbs blue light (1 pt). The absorbed energy twists the molecule into a shape that dissolves in water and leaves in urine and bile without needing the liver's enzyme (1 pt).

7. What did the nurse Sister Jean Ward notice in 1956 that started it all? (1 pt)

Answer: Jaundiced babies she'd wheeled into the sun were paler where the sunlight reached and still yellow under the diaper and sheet folds.

Part E — Flashbacks

8. From NICU Tutor: why are preemies on extra oxygen kept around 90-95% saturation instead of 100%? (1 pt)

Answer: Too much oxygen damages developing eyes (ROP) and lungs; too little is dangerous too. Oxygen is a medicine with a dose.

9. From last week's Physics: a CPAP is set at 6 cmH₂O. About how many mmHg is that? (1 mmHg ~ 1.36 cmH₂O.) (1 pt)

Answer: About 4 mmHg ($6 \div 1.36 \sim 4.4$). A whisper compared with blood pressure.

Part F — Think like a nurse

10. A preemie's pulse ox suddenly reads 75%. The baby is pink, wiggling, and her heart rate on the monitor is a steady 145. You look at her foot: the band has slid sideways and its little red light is shining onto the mattress. Explain, using today's physics, why the number is wrong — and what you'd do first, second, and third. (3 pts)

Answer: The detector is now catching light that never went through blood (mattress, room light), so the red-to-infrared comparison means nothing; a wiggling foot also makes the signal 'pulse' for non-heartbeat reasons (1 pt). First: look at the baby — pink, active, normal heart rate says she's fine (1 pt). Second: reposition the band so the lights face the detector through the foot, shade it from room light, wait for a clean pulse wave; third: if the number is still low with a good signal, then believe it and act (1 pt). Machines report; humans judge.

HOME LAB (optional, not graded): Your finger is a pulse oximeter

You need: a small flashlight or phone light, a dark room, and (bonus) scraps of red and blue cellophane or candy wrappers

1. In the dark, press the flashlight against the pad of your fingertip so it shines INTO the finger.
2. Look at the other side of the finger. What color is the glow? White went in – what came out?
3. Bonus: hold red cellophane over the light and try again; then blue. Which color gets through your finger?
4. Bonus 2: tilt a CD or DVD under a lamp and find the rainbow. Which color is on the outside edge?

What you should see: The fingertip glows orange-red: blood and flesh absorb blue and green and let red pass – the pulse ox's physics. Red cellophane light gets through; blue barely does. The CD splits white light into a spectrum, red on one edge and violet on the other.